

BILL OF MATERIALS

ID	Name	Designator	Footprint	Quantity	Manufacturer Part	Manufacturer	Supplier	Supplier Part
1	Buzzer	B1	Buzzer-12mm SMD	1				
2	CST816	CN1	QFN20-3×3	1	CST816-S	Hynitron Microelectronics	LCSC	
3	GMI6027P-2C30DB	MIC1	MIC-TH_BD6.0-P1.90-D0.4-L	1	GMI6027P-2C30DB	INGHAI	LCSC	C233898
4	ADIA7670C	MOD2	ADIA7670C	1	ADIA7670C	ADIA7670C	Rajguru	ADIA7670C
5	Speaker	S1	SMD 3 solder	1				
6	GC9A01	U1	GC9A01 TFT LCD screen	1	GC9A01	GoldenMorning display		
7	IndusBoard	U2	IndusBoard Coin	1	IndusBoard Coin	IndusBoard	IndusBoard	Coin
8	Antenna	U4	Antenna	1				

```

36 drawHomeScreen();
37 }
38
39 void drawHomeScreen() {
40     tft.fillScreen(TFT_BLACK);
41     tft.setTextColor(TFT_WHITE, TFT_BLACK);
42
43     int buttonWidth = 80, buttonHeight = 60;
44     String apps[4] = {"Contacts", "Calculator", "Games", "Timer"};
45
46     // Display buttons for different apps in a grid
47     for (int i = 0; i < 4; i++) {
48         int x = (i % 2) * (buttonWidth + 10) + 10;
49         int y = (i / 2) * (buttonHeight + 10) + 30;
50
51         tft.drawRoundRect(x, y, buttonWidth, buttonHeight, 5, TFT_WHITE);
52         tft.setCursor(x + 10, y + 20);
53         tft.print(apps[i]);
54     }
55 }
56

```

Fig. 4: Snippet of home screen function to be called by RTOS

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85 tft.print(text + " ");
86 }
87
88 void drawContacts() {
89     tft.fillScreen(TFT_BLACK);
90
91     // Display each contact with a separator line
92     for (int i = 0; i < 3; i++) {
93         tft.setCursor(20, 40 + i * 40);
94         tft.setTextColor(TFT_WHITE, TFT_BLACK);
95         tft.print(contacts[i][0]);
96
97         tft.setCursor(20, 60 + i * 40);
98         tft.print(contacts[i][1]);
99
100        tft.drawLine(0, 80 + i * 40, 240, 80 + i * 40, TFT_WHITE); // Draw line separator
101    }
102 }
103
104 void loop() {
105     if (touch.available()) {

```

Fig. 5: Snippet of contacts function to be called by RTOS

calculator application.

Fig. 2 illustrates the 4G GSM module (left) and MIC (right). The MIC used in this device is an electret microphone capsule, which is commonly used in small electronic devices like mobile phones, hearing aids, and various audio recording devices. These microphones are compact, sensitive, and ideal for small form-factor applications. The tiny holes on top allow sound to enter, where a diaphragm inside the capsule converts sound waves into an electrical signal. The components required to build the IndusPhone are listed in the table above.

Circuit diagram and working

Fig. 3 presents the final circuit diagram for the world's smallest programmable IndusPhone device. It is built around the 4G GSM module (A7670C), self-capacitance touch chip (CST816), microphone (GMI6027P), SMD speaker, LCD screen (FC9A01), antenna, and display. The entire system is powered by a 3.3V battery. The buzzer will ring when an incoming call is detected and serves as an input to the IndusBoard for interrupts related to the incoming call.

The 0.5W SMD speaker is used for both calls and loudspeaker functionality. The condenser mic,